

ECON345 Homework 3: Summary Statistics

Due: September 17th, 11:59 PM

Answer the questions in an R script and upload it to Canvas. Comment your code to explain what you're doing – partial credit exists for a reasonable, well-documented attempt, not just a correct final answer. Some questions have written components; write those answers as comments beneath the relevant code.

Part I: Car Shopping with `mtcars` (9 pts)

We'll use the built-in `mtcars` dataset to practice summary statistics and dataset structure. `mtcars` ships with R – you do **not** need to import a CSV to use it.

Question 1 (2 pts)

Dimensions. Using a single function, report how many rows (cars) and columns (variables) are in `mtcars`.

Question 2 (2 pts)

Structure. Use one function to list the variable names and their data types, and display the result. Are these variables continuous, discrete, or categorical?

Question 3 (5 pts)

Summary statistics. Using a **single** function, report the minimum, maximum, mean, median, and first and third quartiles for the numeric variables in `mtcars`.

1. Using what you know about the relationship between the mean and the median, do you think the distribution of `hp` (horsepower) is skewed or roughly symmetric? If skewed, in which direction?
2. Open the `mtcars` documentation with `?mtcars`. What does `hp` measure, and what kind of car would you expect at the high end of that range?

Part II: California School Districts (11 pts)

Now we'll use the `CASchools` dataset from the **AER** package. Because it comes from a package, you must load **AER** and then run `data("CASchools")` before using it. Use `?CASchools` to read the documentation.

Question 4 (3 pts)

Create a variable. Use `mutate()` to add a variable `stu_teach_ratio`, the number of students per teacher (`students / teachers`). Save the result to a new data frame

CASchools_STR. Then, with a single function, display the mean, median, range, and first and third quartiles of `stu_teach_ratio`.

Question 5 (4 pts)

Bin a continuous variable. `stu_teach_ratio` is continuous, so we can't group by it directly. Using `case_when()` inside `mutate()`, create a categorical variable `str_level` with three levels:

- "Low" — ratio below the first quartile
- "Moderate" — ratio at or above the first quartile and below the third quartile
- "High" — ratio at or above the third quartile

Save it back to `CASchools_STR`. (Hint: compute the quartiles with `quantile()` first.)

Question 6 (3 pts)

Conditional summary statistics. Using `group_by()` and `summarize()`, compute the mean and standard deviation of both `read` and `math` scores for each `str_level`. Save the result to a new data frame.

Question 7 (1 pt)

Interpret. In 2–3 sentences: which student-teacher ratio group has the highest average math score? Which group has the *least* variation in average reading score? Does more crowding (a higher ratio) appear to line up with lower scores?

Submission

Upload your `.R` script to Canvas before the deadline. As always: comment your code, and if you get stuck, show your work.